TECHCIRCUIT HACKATHON · ROUND 1 SUBMISSION · THE PLAN

CanonGate

Pre-publication brand & claim governance platform

Team __Decisionsparalysis__ — Aritra Mahanty · Pulkit Agrawal · Tejas Ranjeet · Kabir Roy

THE PROBLEM

Regulated D2C brands (personal care, F&B, nutraceuticals) and financial-services firms (fintech, NBFCs, digital lending) publish through internal teams, agencies and AI tools. Under deadline pressure, compliance has become post-publication damage control, and most tools either read text only or skip the fine print inside banner images.

11,581
ASCI complaint cases in FY2025-26, up 21% year on year

95.9%
of 1M home pages fail WCAG 2.2 A/AA; low contrast is the top failure (WebAIM 2026)

52 of 53
Indian apps audited by ASCI used dark patterns

₹10L–₹50L
CCPA penalty for a first, then a repeat, misleading-ad offence

OUR SOLUTION

CanonGate compiles brand guidelines, approved facts and statutory rules into a signed, machine-readable Canon, then reviews every campaign bundle against it before publication. Each finding explains itself, points to the exact text span or banner box and the rule involved, and proposes a minimal fix. Humans keep every approval decision.

Tier 1 · Deterministic math & vision
zero AI, runs first

Tier 2 · Calibrated slot-level NLI
runs second, in parallel

Tier 3 · Constrained rewriter
last, flagged spans only

Human sign-off
role-gated, hash-chained

THREE NAMED IMPROVEMENTS WE COMMIT TO DELIVER (ROUNDS 2 AND 3)

1. **Closed-loop Guideline Friction → Canon PR generator:** when a rule's Friction(R_i) exceeds 0.40, auto-draft a Canon v1.1 clarification and simulate its impact on historical bundles.
2. **MCP server and pre-publish gate:** live pre-checks inside Claude/Cursor (MCP 2026-07-28 spec) plus a Shopify/Webflow pre-publish webhook.
3. **Multi-page PDF brochure layout and Hindi-English disclaimer-parity verifier** for code-mixed disclaimers under the 2022 CCPA guidelines.

Round 1 criterion	Problem understanding · 25	Proposed solution · 25	Technical plan · 20	Originality · 15	Feasibility · 15
Where to find it	§1 evidence, §2 users	§3 workflow and architecture	§4 benchmarks and stack	§5 differentiation, §6 responsible design	§7 plan, team, impact

1 Problem definition and evidence

In the ASCI Annual Complaints Report FY2025-26 (28 May 2026), ASCI Chairman Sudhanshu Vats (MD, Pidilite) warned that a “speed-first, compliance-later” culture has normalised “exaggerated claims, manufactured scientific credibility, influencer-led amplification and non-compliance as a post-publication correction exercise” (Exchange4media; ASCI report). Without a pre-publication gate, multimodal bundles (documents plus banners) fail on six dimensions:

1. Qualifier-stripped or unsubstantiated claims
2. Visually subordinated or mismatched disclaimers
3. Expired offers or unverified loan terms
4. Off-brand tone and banned terms
5. Palette drift and logo clear-space intrusion
6. WCAG 2.1 AA contrast and alt-text failures

Evidence source	Verified empirical and statutory finding	What it means for the product
ASCI FY2025-26 report (BuzzInContent, Exchange4media)	• 11,581 cases (+21% YoY) and 9,841 ads (+37% YoY; 98% modified; 93% proactive). Of 9,357 digital ads, 82% were on social and 18% on websites; Meta accounted for 79.84%. • **Personal Care and F&B:** 639 Personal Care cases (90% modified; 45% influencer-led) cited ingredient studies as whole-product proof ("hair growth of 23,800 strands"). 376 F&B cases (88% modified); nutraceuticals were 52% of F&B, with 96% modified.	Scope note: 6,933 cases (~60%) were illegal offshore betting. Legitimate D2C brands fail on ingredient-vs-product substantiation (`substantiation_level`) and missing creator disclosures (76% non-compliance among Forbes Top 100 Digital Stars).
CCPA 2022 Misleading Ads Rules (PIB 1832906, PIB 2238159)	• Consumer Protection Act, 2019 (Sec. 21): penalties up to ₹10 lakh (first offence) and ₹50 lakh (subsequent), 1–3 year endorser bans, and Clause 13 endorsement due diligence. • **Disclaimer parity:** a disclaimer cannot contradict the main claim and "shall be in the same language as the claim... and the font used in a disclaimer shall be the same as that used in the claim."	Requires multimodal verification: text-only tools cannot tell whether a raster banner's disclaimer matches the headline's script, typeface and visual prominence.
RBI Digital Lending 2025 and SEBI AI rules (RBI/2025-26/36, AZB / SEBI)	• RBI Directions (8 May 2025) consolidate digital-lending rules (repealing the 2020, 2022 Digital Lending and 2023 DLG circulars in Annex III, and incorporating the 15 April 2024 KFS circular DOR.STR.REC.13/13.03.00/2024-25 under Para 4(i) and 8(i)), mandate APR, penal charges and KFS disclosure, and ban dark patterns (Para 6(iv)). • SEBI (6 Feb 2025) assigns sole legal liability to regulated entities for "the output resulting from the usage of such [AI and ML] tools and techniques."	Financial promos must be checked against structured KFS fee/APR schedules and campaign `promo_valid_until` dates, backed by an immutable human sign-off log.
WebAIM 2026 and industry surveys (WebAIM, CMI, Gartner, ASCI)	• WebAIM 2026 (1M pages): 56,114,377 errors (56.1 per page, +10.1% YoY); 95.9% fail WCAG 2.2 A/AA. Low contrast hit 83.9% of pages (34 per page); 27.0% of 66.6M images had missing or defective alt text; `.in` domains averaged 58.3 errors. • CMI (N = 1,015) and Gartner (N = 401): 95% use AI (89% for copy, 53% for visuals), yet 68–70% lack mature governance processes to scale it. • ASCI Conscious Patterns (53 Indian apps): 52 of 53 used dark patterns (Interface Interference 45.3%, 55% in fintech; Drip Pricing 43.4%).	WebAIM's WAVE engine inspects the HTML DOM only, so it is blind to text rasterised inside PNG/JPEG banners. CanonGate audits both HTML/DOCX alt text and banner pixel luminance.

1.1 Adjudicated CCPA orders and which CanonGate check would have caught each

₹10 LAKH · AUG 2025

Rapido (PIB 2158830)

Ran ads across 120+ cities for ~548 days claiming "AUTO IN 5 MIN OR GET ₹50" while rendering "T&C Apply" in unreadable font and omitting four qualifiers (up to ₹50, paid in Rapido coins, bike rides only, 7-day expiry).

CAUGHT BY: Tier 2 SlotDiff (dropped qualifiers) + Tier 1 disclaimer prominence parity.

₹15 LAKH · FEB 2026

Vajirao & Reddy (PIB 2231796)

Claimed "Over 645 Selections Out of 1016 Vacancies in UPSC CSE 2023" beside Foundation courses without disclosing most took only Mock Interviews (431 forms lacked course names/dates). Fined after a prior ₹7L CSE 2022 penalty, among 57 CCPA notices and ₹1.246 Cr across 29 institutes.

CAUGHT BY: Tier 2 SlotDiff on the claim's context and `open_conditions[]`.

₹2 LAKH · SEPT 2025

FirstCry (PIB 2171807)

Paired "MRP inclusive of all taxes" with footnotes stating "GST and Add'l charges may apply on discounted price," dropping a 27% discount to 18.2% at checkout.

CAUGHT BY: Tier 2 cross-modal contradiction between headline and footnote.

> ***"When accountability is shared by everyone, it often ends up being owned by no one."***
> Vibhor Gulati and Ajay Kulkarni, IMPACT, 15 July 2026 (source). ASCI CEO Manisha Kapoor makes the same point on outcome scrutiny (Adgully, 18 Dec 2025) and ingredient-vs-product claims (Forbes India, 16 Sept 2026).

EVIDENCE STATUS AND PRIMARY-RESEARCH PLAN

Everything above is secondary evidence: regulator reports, adjudicated orders, industry surveys and press interviews. Primary validation is scheduled for Round 2:

- **5–8 practitioner sessions** across brand managers, compliance/legal reviewers and agency creators in D2C and digital lending, led by Kabir Roy.
- **What we will test:** whether the Blocker/Warning split matches how they triage, whether the SlotDiff card is understandable without training, and which Canon rules they find ambiguous.
- Findings will feed the Friction(R_i) analytics and be reported in the Round 2 submission.

2 Target users, needs and organisational setting

Setting. A marketing team at a mid-market Indian D2C or BFSI company publishes social posts, web pages, brochures and banners through in-house staff, agencies and AI tools. A brand manager owns the guidelines, an authorised legal or compliance reviewer holds regulatory sign-off, and the regulated entity carries the liability (SEBI makes this explicit for AI-generated output). Review today tends to happen late, and accountability is spread so thin that no one owns it.

ROLE 1 · THE CREATOR

NEEDS: Fast, specific feedback before publishing, without losing the point of the copy.

CANONGATE GIVES:

- Blockers and Warnings, priority-grouped
- Exact `[start, end]` text and `[x, y, w, h]` banner highlights
- One-click surgical diffs and inline comments
- A Deliberate Override button for Warnings

ROLE 2 · THE BRAND MANAGER

NEEDS: Turn a PDF brand book into enforceable rules quickly, and learn which rules confuse people.

CANONGATE GIVES:

- 5-minute Compile-to-Schema onboarding with Confirm & Calibrate sliders
- PR-style diff and revision history
- Guideline Friction dashboard

ROLE 3 · THE AUTHORISED LEGAL REVIEWER

NEEDS: Review only what is genuinely at risk, with a defensible record.

CANONGATE GIVES:

- Claim-only filter with a structured SlotDiff table
- Hard role gate
- Append-only SHA-256 audit certificate

3 Proposed solution and end-to-end workflow

Value proposition: CanonGate is a self-hostable, multimodal pre-publication gate that catches qualifier-stripped claims, mismatched disclaimers, expired offers, off-brand language and inaccessible banners before publication, explains each finding against the approved source, and leaves every approval to a human. It follows one strict execution order: deterministic math and SQL first, calibrated NLI second, constrained rewriting last.

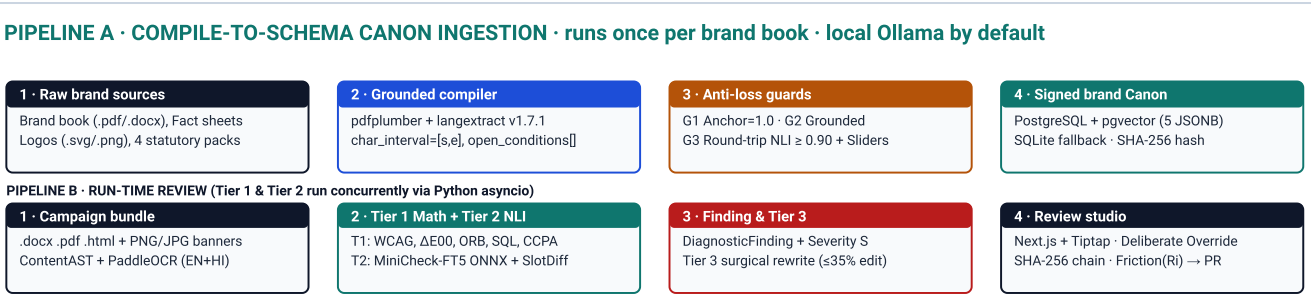


Figure 1. Two-pipeline architecture. Pipeline A compiles brand sources into five JSONB schemas behind three anti-loss guards. Pipeline B parses campaign bundles into one ContentAST, runs Tier 1 and Tier 2 in parallel, invokes Tier 3 only on flagged spans, and routes findings to the three-role review studio.

3.1 Four-stage collaborative workflow

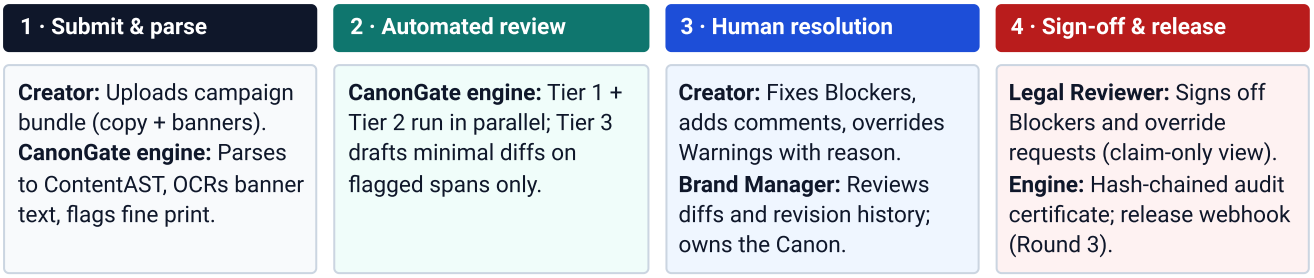


Figure 2. Who does what, and when. Warnings can be overridden by the Creator with a recorded reason; Blockers need a fix or Authorised Reviewer sign-off. Every step is logged in the hash-chained audit trail.

Stage	Who	What happens
1. Submit & parse	Creator	Uploads a bundle (`.docx .pdf .html .md .png .jpg`). It becomes a unified `ContentAST` of `TextNodes` and `VisualNodes` via PaddleOCR 3.x (PP-OCRv5); Hindi/Devanagari banners route to a dedicated recogniser, and any fine-print box with OCR confidence below 0.75 or height under 10 px is flagged for human review.

2. Automated review	CanonGate engine	Tier 1 and Tier 2 run concurrently; Tier 3 proposes fixes only on flagged spans and every candidate is re-verified by Tiers 1 and 2 before display. Each result is a `DiagnosticFinding` JSON (`SourceLocator`, `SlotDiff`, `ErrorDegree`).
3. Human resolution	Creator, Brand Manager	Creator accepts or edits surgical diffs, comments inline, and may deliberately override Warnings. Brand Manager reviews the revision diff and owns the Canon.
4. Sign-off & release	Authorised Legal Reviewer	Reviewer sees only unresolved Blockers and override requests. Approval writes a SHA-256 hash-chained certificate; a CMS pre-publish webhook releases the content (Round 3).

3.2 What a finding looks like

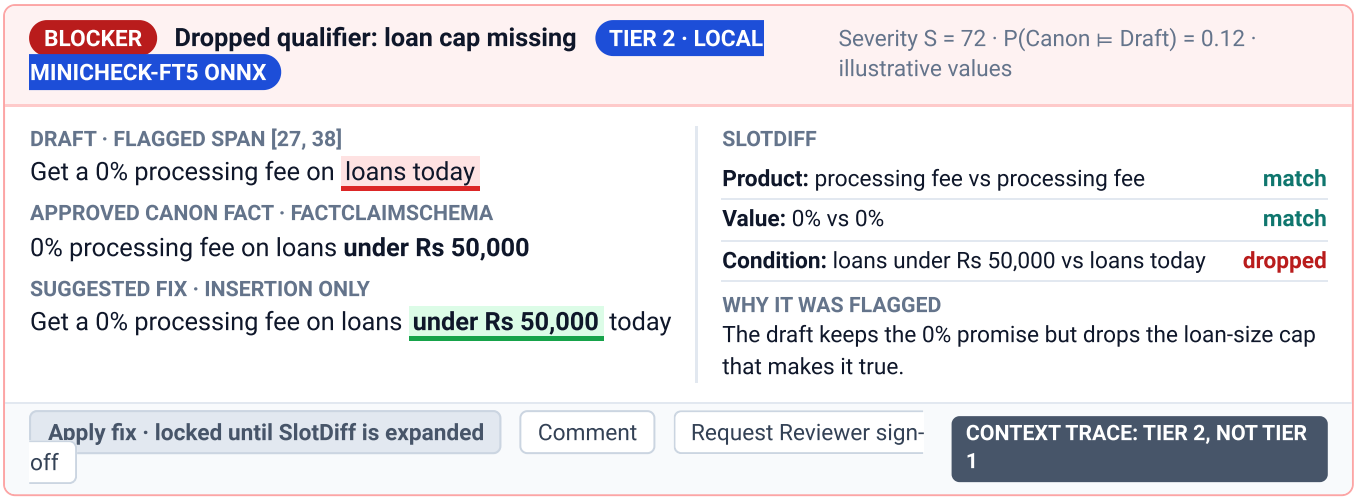


Figure 3. Illustrative finding card (values are examples, not measurements). The same canon claim is used in Benchmark 1. The locked button is one of the cognitive forcing functions in §6 that stop reviewers rubber-stamping Blockers.

3.3 Pipeline A: compile-to-schema, and three guards against silent rule loss

Uploaded brand guidelines (`.pdf/.docx`), product/KFS fact sheets (`.csv/.json`) and reference logos (`.svg/.png`) compile into five PostgreSQL 16 + JSONB schemas:

Schema	Holds
`FactClaimSchema`	`subject_product`, `metric_value`, `comparison_operator`, `promo_valid_until`, `kfs_acceptance_validity_days`, `substantiation_level` (`finished_product_clinical` vs `ingredient_in_vitro`), and a JSONB `open_conditions[]` array inspired by zero-shot open-schema discovery in ZOES (Xu et al., EACL 2026).
`TriggerObligationSchema`	`IF trigger_pattern THEN required_disclaimer_id`, with `same_language` and `same_font_family` flags and a configurable `min_font_height_ratio` for prominence.
`TerminologySchema`	Banned terms, preferred replacements, part-of-speech constraints.
`VoiceProfileSchema`	Sentence-length bounds, maximum passive-voice %, Flesch-Kincaid grade, and `pgvector` embeddings of approved brand-voice exemplars.
`VisualTokenSchema`	Approved hex codes with CIE LAB coordinates and a maximum ΔE_{00} tolerance, logo ORB keypoints, clear-space ratio (default 0.5H), and WCAG 2.1 AA thresholds.

Because `langextract` (v1.7.1) uses JSON-mode output rather than grammar-enforced schemas on local Ollama, three guards prevent silent condition dropouts:

GUARD 1 · VALUE-ANCHOR COVERAGE AUDIT

spaCy extracts every number, %, amount, duration and restrictive modifier (*only, under, up to, within*) from sentence S and enforces |Anchors(JSON)| / |Anchors(S)| = 1.0, re-querying any uncaptured condition into `open_conditions[]`.

GUARD 2 · VERBATIM CHARACTER GROUNDING

`langextract` maps each rule to its exact source span (`char_interval = [start, end]`) using local Ollama (default for confidential canons) or opt-in `gemini-3.5-flash` with two passes. Ungrounded extractions return `None` and are rejected.

GUARD 3 · ROUND-TRIP NLI CHECK

Each JSON rule is rendered back to prose S_{recon}; local MiniCheck-FT5 must find entailment in both directions, $\min(P(S \models S_{recon}), P(S_{recon} \models S)) \geq 0.90$, alongside the Confirm & Calibrate sliders for the Brand Manager.

3.4 Pipeline B: the three tiers

Tier	What it checks	Why this order
Tier 1 Deterministic, zero AI	spaCy POS terminology; `textstat` readability; SQL checks on `promo_valid_until` and numeric caps; W3C WCAG 2.1 pixel-luminance contrast; `skimage` CIEDE2000 palette distance; OpenCV ORB logo aspect distortion (≤ 2%) and 0.5H clear-space intrusion; CCPA headline-vs-disclaimer language and prominence parity.	Exact, repeatable and explainable: a rule either holds or it does not.
Tier 2 Calibrated NLI + SlotDiff	Retrieves the top-3 Canon facts via `pgvector` + BM25; scores directional entailment P(Canon ⊨ Draft) with local MiniCheck-Flan-T5-Large (770M), calibrated by temperature scaling; aligns predicate-argument slots with QASem and spaCy anchors to fill the SlotDiff table, exposing dropped qualifiers, ingredient-vs-product mismatches and cross-modal contradictions.	Meaning, not wording, decides whether a claim is supported, and borderline scores go to a person.
Tier 3 Constrained rewriter	Invoked only on flagged spans. Schema-constrained generation (local Ollama, or opt-in `gemini-3.5-flash`) under a `rapidfuzz` Levenshtein ceiling (≤ 35% edit distance outside inserted qualifiers); every candidate is re-verified by Tiers 1 and 2 before it is shown.	Preserves the creator's intent; never overrides a human decision.

4 Technical plan and reproducible benchmarks

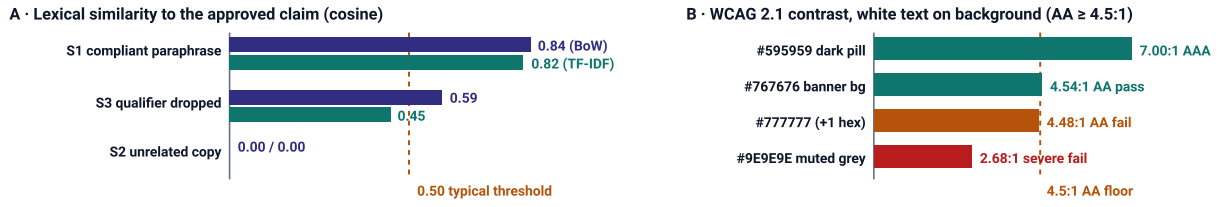


Figure 4. Why the architecture is built this way. Left: lexical similarity of three drafts to the approved claim C0. Right: exact W3C relative-luminance contrast. Both were re-computed in Python for this submission (bag-of-words uses a lowercase `[a-z0-9]+` tokenizer; TF-IDF is scikit-learn defaults).

Benchmark 1: lexical similarity measures overlap, not entailment. Approved fintech claim C_0 = “0% processing fee on loans under Rs 50,000”. A compliant paraphrase S_1 scores 0.84 (bag-of-words) and 0.82 (TF-IDF); an unrelated

copy S_2 scores 0.00 on both; the qualifier-stripped violation S_3 ("Get a 0% processing fee on loans today") still scores 0.59 and 0.45, right at the 0.50 range typically treated as "related". Dropping a three-word cap leaves the core nouns intact, and cosine is symmetric, so it cannot say which direction the claim is supported; any threshold strict enough to reject S_3 also rejects legitimate paraphrases with different wording. That is why Tier 2 uses directional entailment plus SlotDiff alignment.

Benchmark 2: a one-hex-unit shift flips WCAG compliance. Under WCAG 2.1 Technique G18 (L = 0.2126R + 0.7152G + 0.0722B; C = $(L_1 + 0.05)/(L_2 + 0.05)$), white text ($L_1$ = 1.0000) on `#767676` (L_2 = 0.1812) gives 4.54 : 1 (passes AA), while `#777777` (L_2 = 0.1845) gives 4.48 : 1 (fails AA). A muted footnote grey `#9E9E9E` (L_2 = 0.3419) gives 2.68 : 1 (severe fail), and CanonGate's suggested dark pill `#595959` (L_2 = 0.0999) gives 7.00 : 1 (passes AAA).

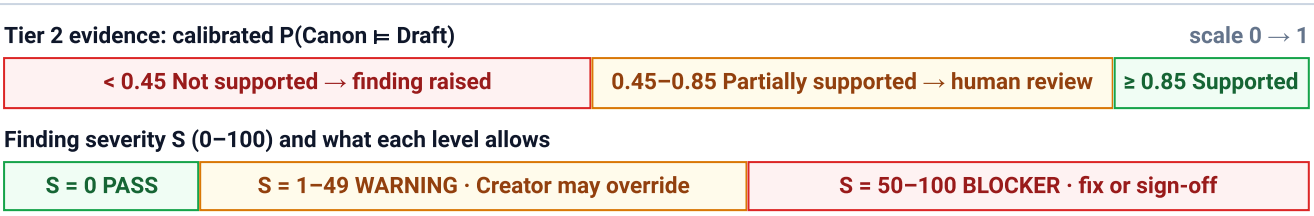


Figure 5. Calibrated uncertainty and severity routing. Borderline entailment scores are never guessed: they surface as "Partially Supported — Needs Human Review".

4.1 Technology stack, licences and owners

Layer & owner	Technology & licence	Rationale and fallback
Studio and canvas Tejas Ranjeet	Next.js 15 (React 19, TypeScript), Tiptap headless editor, HTML5 canvas / SVG overlay, Recharts (all MIT)	Tiptap's ProseMirror AST maps `[char_start, char_end]` to surgical-diff decorations, comment threads and revision history; SVG overlays render `[x, y, w, h]` banner flags and the calibration sliders.
Ingestion and guards Aritra Mahanty, Pulkit Agrawal	`google/langextract` v1.7.1 (Apache-2.0; local Ollama default, opt-in `gemini-3.5-flash`), `pdfplumber` (MIT), `pypdf` (BSD-3), `python-docx` (MIT)	Char-interval provenance and multi-pass recall, paired with Guards 1–3. MIT/BSD parsers avoid PyMuPDF's AGPLv3 copyleft restrictions.
Tier 1: text, vision, accessibility Aritra Mahanty	spaCy, `textstat`, `langdetect` (MIT); OpenCV (Apache-2.0); `scikit-image` (BSD-3); PaddleOCR 3.x PP-OCRv5 (Apache-2.0)	English recogniser (85.25%) and Devanagari recogniser (84.96%, +68.26% over v3) with confidence gating; `deltaE_ciede2000` for exact ΔE_{00} drift; ORB for logo aspect and 0.5H clear space.
Tier 2: fact and slot entailment Aritra Mahanty	MiniCheck-Flan-T5-Large 770M (weights MIT, paper CC-BY-4.0), MiniCheck-DeBERTa-v3-Large 0.4B fallback, QASem (MIT), ONNX Runtime (MIT)	Per Tang et al., EMNLP 2024 (Table 2), MiniCheck-FT5 reaches 74.7% balanced accuracy against GPT-4's 75.3% on LLM-AggreFact (arXiv:2404.10774v2) at ~400x lower cost. If CPU ONNX latency exceeds 400 ms, auto-fall back to the DeBERTa model.
Tier 3, storage, security, connectors Pulkit Agrawal, Kabir Roy	Ollama (`gemma2:2b/qwen2.5:7b`) or opt-in `gemini-3.5-flash`; `rapidfuzz` (MIT); FastAPI (MIT); PostgreSQL 16 + `pgvector`, SQLite fallback; MCP server (2026-07-28 spec) and REST webhooks	Stores the five Canon schemas, comment threads, version history and the append-only SHA-256 audit log, with a zero-config SQLite mode so judges can run it without setup. Round 3 adds `canongate-mcp-server` and CMS pre-publish webhooks.

Data sources shipped with the product. Three starter Canons: (1) B2B SaaS and design brand specification (palette tokens, typography, 0.5H logo clear space, active-voice bounds); (2) regulated Indian digital lending (loan caps ≤ ₹50,000, tenure rules, `promo_valid_until`, RBI 2025 KFS APR/fee schedules); (3) regulated Indian D2C personal care and nutraceutical (`ingredient_in_vitro` vs `finished_product_clinical` per ASCI FY2025-26). Four statutory packs (CCPA 2022, RBI 2025 KFS, ASCI Influencer, WCAG 2.1 AA) and the CanonGate-50 red-team suite, which reconstructs the Rapido, FirstCry and Vajirao & Reddy orders plus boundary WCAG contrast cases.

5 Originality and competitive differentiation

CanonGate differs from DAM suites and text checkers on five axes:

1	2	3	4	5
Pre-compiled Indian statutory packs (CCPA, RBI KFS, ASCI)	Slot-level claim entailment with character and box provenance	Deterministic physical & linguistic math (WCAG, ΔE_{00}, ORB, POS)	Closed-loop friction analytics: Friction(R_i) = (Overrides + Disputes) / Flags	Self-hostable, local-first container deployment

Platform (2026)	Verified capabilities	Documented limitations and how CanonGate differs
Bynder Brand Compliance Agent (press, docs) and **Adobe Brand Intelligence** (Adobe)	Bynder audits DAM assets for logo placement, clear space, distortion, palette, typography and sector rules. Adobe validates images and copy against a brand ontology with visual issue pins and decision traces.	**Fine-print and downscaling blind spots:** Bynder's docs say text under ~0.15% of image area may be skipped, colour matching runs on a downscaled copy, and agents cap at 50 rules / 10K characters. Neither is self-hostable or ships Indian statutory packs.
Frontify Specs (Frontify) and **Marq Brand Guardian** (Marq)	Frontify Specs is a vision-language agent (preview, Feb 2026) checking logos, fonts, colours and tone. Marq offers lockable templates, Smart Fields for regional disclosures and drift checks.	**Template and VLM reliance:** Marq needs assets authored inside locked templates rather than auditing external agency PNGs/DOCXs; Frontify Specs is in preview and uses probabilistic VLMs, not deterministic WCAG luminance or CIEDE2000 math.
Markup AI (docs, pricing, Oct 2026 blog)	Priority-grouped sidebar addressing "red-ink overwhelm", persona simulation, Google Docs/Slides and Word add-ons, Claude/ChatGPT MCP ($249/mo Starter, $999/mo Business).	**Text-only sidebar:** Slides were added alongside Docs in Oct 2026, but the docs state that images, charts and other non-text elements are not analysed. Custom criteria and Figma need Enterprise pricing.
Writer (eesel 2026), **CustomGPT** (Verify), **Sedric** (Sedric)	Writer: knowledge-graph citations ($29–$39/user/mo Starter, 5-seat cap). CustomGPT: three-tier claim badges and six-perspective risk review. Sedric: FinServ policy grid (UDAAP, Reg Z, FINRA) and ad scanning.	**Setup friction and feature gates:** across 105 G2 reviews (4.3/5), Writer users cite a steep learning curve at setup, and Starter excludes audit logs, RBAC, SSO and cross-team workflows. CustomGPT checks only chatbot Q&A; Sedric lacks visual brand math.

6 Responsible design and human oversight

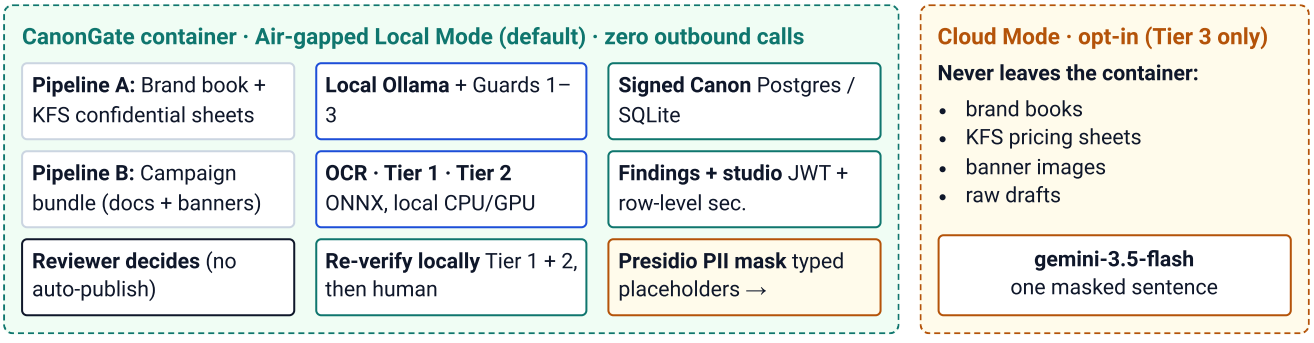


Figure 6. Trust boundary. In the default Local Mode nothing leaves the container. In opt-in Cloud Mode, only one PII-masked flagged sentence is sent for a Tier 3 rewrite; the candidate is re-verified locally and a human decides. The use of AI and any external service is disclosed in every finding.

GLASS BOX · INTERPRETABLE, CALIBRATED, HONEST ABOUT DOUBT

Tier 1 rules are 100% deterministic with zero neural weights. Tier 2 shows the SlotDiff table beside the temperature-calibrated P(Canon ⊨ Draft) and routes borderline scores ($0.45 \le P < 0.85$) to "Partially Supported — Needs Human Review" instead of guessing.

SECURITY · HARDENED AGAINST THE OWASP TOP 10 FOR LLM APPS (2025)

Against LLM01 indirect prompt injection (for example 1 px white-on-white "Ignore all rules"), Tiers 1 and 2 are non-generative and never execute text as instructions, while a DOM style inspector and OpenCV pixel-gradient scanner flag hidden text ($1.00 : 1 < 4.50 : 1$) as a Blocker. Against LLM06 excessive agency, Tier 3 has no tool-calling or DB write permission; only authenticated reviewers (JWT + PostgreSQL row-level security) can approve a release (OWASP).

PRIVACY · LOCAL-FIRST, ALIGNED WITH INDIA'S DPDP RULES, 2025

Prelaunch campaigns and KFS pricing sheets are trade secrets, so Pipeline A, parsing, OCR, Tier 1 and Tier 2 ONNX run locally with zero outbound calls. In Cloud Mode, local Microsoft Presidio plus custom regex mask PII, phone numbers, emails, PAN/Aadhaar and project codenames into typed placeholders. Ephemeral Audit Mode (`RETENTION_DAYS=0`) keeps only SHA-256 hashes and diagnostic metadata; tenants are isolated and audit logs are in plain language (PIB 2190014).

HUMAN OVERSIGHT · FORCING FUNCTIONS AGAINST RUBBER-STAMPING

Grounded in Buçinca et al., CSCW 2021 (N = 199): (1) no bulk "Accept All" on Blockers, since the SlotDiff card must be expanded before action buttons unlock; (2) a structured Deliberate Override popover; (3) a mandatory Context Trace badge saying whether a finding came from Tier 1 (zero AI), Tier 2 (local MiniCheck-FT5 ONNX) or Tier 3 (surgical rewriter).

Tamper-evident audit chain. $H_n = \text{SHA256}(H_{n-1} \,||\, \text{timestamp}_n \,||\, \text{actor_id}_n \,||\, \text{role}_n \,||\, \text{canon_hash} \,||\, \text{bundle_hash} \,||\, \text{decision}_n)$, append-only, so any edit to history breaks every later hash.

7 Feasibility, development plan, team and impact

ROUND 1 · THE PLAN

Problem 25 · Solution 25 · Tech 20 · Originality 15 · Feasibility 15

- Two-pipeline architecture, 5 JSONB schemas, 3 anti-loss guards
- Reproducible benchmarks: lexical-similarity gap and WCAG boundary
- 3 starter Canons, 4 statutory packs, module ownership

Gate: every statistic links to a source; benchmarks re-run in Python.

ROUND 2 · THE BUILD

End-to-end 30 · Depth 25 · Progress 20 · Improvements named 15 · Usability 10

MUST-HAVE · SPRINT 1 · ENGINE

- Pipeline A (`langextract` + Guards 1–2)
- ContentAST, Tier 1 text/vision/WCAG/CIEDE2000/CCPA engine
- Tier 2 local ONNX MiniCheck-FT5 + SlotDiff
- 3-role approval gate, comments, revisions, SHA-256 certificate

SHOULD-HAVE · SPRINT 2 · POLISH

- Guard 3 + Confirm & Calibrate sliders
- Tier 3 rewriter (≤35% edit, Presidio scrub), ORB logo checks, Live Friction(R_i) chart

Gate: runs live in Docker/SQLite without mocks; ONNX latency > 400 ms → DeBERTa fallback; feedback from 5–8 practitioners.

ROUND 3 · THE FINISH

Completeness 25 · Improvements 20 · Reliability 20 · Fit 20 · Docs 15

- **Improvement #1:** Friction → Canon PR generator (Friction(R_i) > 0.40 triggers a v1.1 draft and impact simulation)
- **Improvement #2:** MCP server + Shopify/Webflow pre-publish gate
- **Improvement #3:** multi-page PDF brochures + Hindi-English disclaimer-parity verifier

Gate: 100+ concurrent bundle runs, OWASP LLM01/LLM06 red-teaming, simulated cloud-API outages.

7.1 Team and module ownership

Member	Role	Owned modules and responsibilities
Aritra Mahanty	Lead ML/AI, multimodal vision and verification architect	`/backend/engine/tier1..3/`, `/backend/ingestion/guards/`, `/benchmarks/`: Pipeline A compiler and three guards; Tier 1 vision and accessibility math (WCAG luminance, CIEDE2000, ORB, dual-script PP-OCRv5); Tier 2 quantised ONNX MiniCheck-FT5, calibration and QASem SlotDiff; Tier 3 rewriter and re-verification loop.
Pulkit Agrawal	Lead backend, database, security and privacy	`/backend/db/`, `/backend/security/`, `/backend/ast/`, `/docker/`: five PostgreSQL + JSONB + `pgvector` schemas with SQLite fallback; `ContentAST` parser; Presidio PII scrubbing and Ephemeral Audit Mode; JWT RBAC state machine; hash-chained certificate generator.
Tejas Ranjeet	Lead frontend, collaborative workflow and human-oversight UX	`/frontend/studio/`, `/frontend/canvas/`, `/frontend/workflow/`: Next.js 15 + Tiptap split-screen studio; SVG banner overlays; calibration sliders; comment threads and revision history; forcing-function UX (forced SlotDiff expansion, Override popover, Context Trace badges).
Kabir Roy	Lead governance analytics, integrations and red-team QA	`/backend/analytics/`, `/integrations/mcp/`, `/datasets/canongate-50/`: Friction(R_i) dashboard and Round 3 Canon PR generator; starter Canons, statutory packs and CanonGate-50 suite; practitioner usability testing; MCP server and CMS webhooks.

7.2 Expected impact and roadmap

IMPACT

- **Deterministic precision on Tier 1 rules:** expired dates, numeric caps, POS-constrained terms, WCAG luminance down to a one-hex shift, missing trigger disclaimers.
- **~400x lower fact-check cost** than GPT-4 using local MiniCheck-FT5 (Tang et al., EMNLP 2024).
- **Faster legal sign-off:** reviewers see only unresolved SlotDiff cards and override requests.

AFTER ROUND 3: PRODUCT TRACKS

- **Open-source core + self-hosted VPC edition** for Indian D2C and BFSI, with ASCI, CCPA, RBI, SEBI and FSSAI packs for mid-market teams priced out of $50k+/yr DAM suites.
- **Figma / Adobe Express and Meta Ads Manager / Shopify plugins** that verify ads and product cards before spend.

AFTER ROUND 3: NEW MEDIA

- **Video and audio reel compliance:** local Whisper speech-to-text plus keyframe OCR to check spoken claim substantiation and upfront visual disclosure labels in creator reels.